

MEENA KANNAN

Lead Software Engineer | Cloud-Native Modernisation | AI-Augmented Engineering | GovTech & Fintech

Singapore & India | +65 87373057 | +91 97893 02084 | meenakannan92@gmail.com
linkedin.com/in/meena-kannan-mk | github.com/meenak95

Citizenship: India | Available from Singapore or India | 9+ Years Experience | Open to Senior Remote Roles Worldwide

PROFESSIONAL SUMMARY

Lead Software Engineer with 9+ years of experience delivering production-grade systems for Singapore government agencies, fintech, legal tech, and enterprise SaaS. Currently leading VRLS 2.0 at LTA, modernising a Java monolith into Kubernetes-based microservices on AWS Government Cloud with Kafka, Istio, KEDA, ArgoCD, Redis, and observability-first delivery. Strong hands-on experience across Java 21, Spring Boot 3, Angular, cloud-native architecture, secure delivery, and AI-assisted engineering using Spring AI, RAG, MCP, Claude API, and GitHub Copilot agentic workflows.

CORE STRENGTHS

Cloud-Native Modernisation Java 21, Spring Boot 3, Kubernetes, Helm, Istio, AWS Government Cloud	AI-Augmented Engineering Spring AI, RAG, MCP, Claude API, Copilot agentic mode	Event-Driven Platforms Kafka, Kafka Streams, KEDA, outbox pattern, saga choreography
DevSecOps & GitOps ArgoCD, Jenkins, GitHub Actions, SonarQube, Fortify, Vault	GovTech Delivery LTA and HDB platforms, UAT ownership, SLA-driven support	Engineering Leadership Scrum lead, ADRs, mentoring, reviews, knowledge transfer

TECHNICAL SKILLS

Languages	Java 21, TypeScript, JavaScript, Python, Groovy, SQL, Bash
Backend	Spring Boot 3.x, Spring Security, Hibernate/JPA, WebFlux, Resilience4j, REST, SOAP
AI & LLMs	Spring AI, RAG pipelines, MCP, Claude API, embeddings, prompt engineering, evaluation loops, tool orchestration
Frontend	Angular 18, NgRx, RxJS, React, Next.js, JSF/PrimeFaces, GraphQL
Cloud & Infra	AWS ECS, EKS, ALB, S3, RDS, CloudFront, API Gateway, Lambda, IAM, Cloudflare, NGINX, Istio, Docker, Kubernetes, Helm
Messaging & Data	Kafka, Kafka Streams, KEDA, JMS, IBM MQ, PostgreSQL, Oracle, MySQL, MongoDB, Redis, Flyway
DevSecOps	Jenkins, GitHub Actions, ArgoCD, Terraform, Ansible, SonarQube, HP Fortify, Vault, Keycloak, OAuth2/OIDC
Observability & Testing	Prometheus, Grafana, OpenTelemetry, ELK, Jaeger, Dynatrace, JUnit, Mockito, Cypress, Selenium, JMeter

AI TRACKS & SPECIALISATIONS

Agentic AI & LLM Integration Using GitHub Copilot agentic workflows in production delivery. Prototyping Spring AI and Claude API patterns for Java service acceleration.	RAG & Retrieval Systems Designing RAG pipelines with vector retrieval, chunking strategy, embeddings, and domain-specific retrieval evaluation.
MCP & Tool-Augmented Agents Building MCP server patterns for backend tool invocation, code-review assistance, and microservice instrumentation.	Responsible AI & Governance Applying PDPA-aware data minimisation, auditability, and evaluation loops to reduce hallucination and production risk.

PROFESSIONAL EXPERIENCE

Lead Software Engineer | NCS Singapore | Client: Land Transport Authority (LTA) Dec 2023 - Present

Promoted from Application Consultant after owning the legacy platform knowledge transfer that unblocked VRLS 2.0.

- Architected the Spring Boot 3 microservices backbone for VRLS 2.0 across vehicle registration, ownership transfer, road tax, and permits, with each service owning its data and deployment lifecycle.
- Implemented Istio service mesh for Kubernetes DNS service discovery, mutual TLS, traffic policy enforcement, and safer service-to-service communication without heavy application-level changes.
- Designed AWS ALB with NGINX Ingress for path-based routing, SSL termination, health checks, and controlled traffic flow into Kubernetes workloads.
- Built Kafka event streaming, KEDA autoscaling, and Redis caching to reduce polling, timeout cascades, and manual scaling during peak renewal periods.
- Led Angular 13 to 18 migration with NgRx and RxJS patterns, reducing redundant API calls by about 40% and improving maintainability.
- Delivered workloads on AWS Government Cloud using ECS, EKS, ALB, S3, RDS, CloudFront, Secrets Manager, and IAM for a citizen-facing platform serving millions of users.
- Standardised secure releases with ArgoCD GitOps, Jenkins, GitHub Actions, SonarQube, and HP Fortify SAST.
- Rolled out GitHub Copilot agentic workflows and prototyped Spring AI, MCP, and Claude API integrations for squad adoption.
- Instituted ADRs, code reviews, junior mentoring, sprint planning, and retrospectives as technical Scrum Lead.

Stack: Java 21, Spring Boot 3, Angular 13 to 18, NgRx, Kafka, Kafka Streams, KEDA, Redis, PostgreSQL, AWS, Cloudflare WAF, NGINX, Istio, ArgoCD, Docker, Kubernetes, Helm, Jenkins, GitHub Actions, Vault, Keycloak, Prometheus, Grafana, OpenTelemetry, Dynatrace, SonarQube, Fortify, Spring AI, GitHub Copilot, MCP

Application Consultant | NCS Singapore | Client: Land Transport Authority (LTA) Dec 2021 - Dec 2023

Designated technical owner of the live VRLS legacy platform and author of the knowledge base that enabled the VRLS 2.0 greenfield build.

- Managed production support for live VRLS using Java, Spring MVC, Oracle DB, WebLogic, Kafka, and JMS while maintaining SLA commitments under GovTech incident-response expectations.
- Produced handover documentation for 8 core licensing modules, enabling the VRLS 2.0 squad to move forward without prior domain knowledge and reducing requirement-clarification cycles.
- Served as the single technical point of contact with LTA through UAT and the parallel-run period, bridging legacy behaviour gaps for the incoming engineering team.

Stack: Java, JSP, Spring MVC, Oracle DB, Kafka, JMS, Redis, REST, WebLogic, JUnit, HP Fortify

Senior Software Engineer | BlackStraw.AI | India

May 2021 - Nov 2021

AI-powered retail analytics platform for CPG trade-promotion forecasting.

- Built Spring Boot microservices connecting retail client data feeds to AI inference endpoints, delivering real-time promotion performance visibility across client portfolios.
- Integrated AI/ML REST APIs with batching, fallback handling, and latency SLAs. Replaced batch polling with Kafka event streams, cutting execution-reporting delays from 15+ minutes to near real time.

Stack: Java, Spring Boot, Groovy, Apache Kafka, AWS Lambda, PostgreSQL, ML REST integration, Jenkins, SonarQube

EARLIER EXPERIENCE

Software Engineer | HDB via AllTech Systems | Singapore

Aug 2019 - Dec 2020

Housing & Development Board citizen-facing property transaction portals.

- Developed JSF/PrimeFaces UI and Java backend features for the Resale One Stop Portal, supporting guided multi-step citizen transactions.
- Built eligibility-check and fee-computation flows for the Lease Buyback System with multi-level approval routing and an auditable state machine for regulatory compliance.

Stack: Java, JSF/PrimeFaces, JSP, Oracle DB, WebLogic, SOAP, JUnit, SVN

Software Engineer | Wolters Kluwer | India

Jul 2018 - Jul 2019

Legal technology products.

- Built the Spring Boot microservice behind Legal View Bill Analyzer, combining AI model output with legal expert input to flag billing anomalies.
- Engineered a Drools BRMS rule engine for multi-jurisdiction legal billing compliance, delivering sub-500 ms rule evaluation under production load.

Stack: Spring Boot, MongoDB, Groovy, Drools, Oracle, REST, Jenkins, SonarQube

Associate Software Engineer | SysArc Infomatix Pvt. Ltd. | India

May 2016 - Jul 2018

Banking and financial technology systems.

- Built the full loan lifecycle backend for LAPS, covering origination, credit decisioning, disbursement, and loan servicing.
- Developed Struts/JSP loan officer workflows, replacing paper-based processes and improving loan processing turnaround.

Stack: J2EE, Spring MVC, Hibernate, Struts, Oracle DB, JUnit, SVN

CERTIFICATIONS

Agile & Professional	• Professional Scrum Master I (PSM I), Scrum.org • Professional Scrum Product Owner I (PSPO I), Scrum.org • ICT Assessment Certification, NUS-ISS Singapore
AI, LLM & Cloud	• AI Fluency Framework & Foundations, AI Singapore / AIXTech • Building with the Claude API, Anthropic • Model Context Protocol: Advanced Topics, Anthropic • AWS Educate Machine Learning Foundations, Amazon Web Services • AWS Educate Introduction to Cloud 101, Amazon Web Services
Targeted Upcoming Certifications	• AWS Certified Machine Learning Engineer - Associate • AWS Certified AI Practitioner • Google Cloud Generative AI Leader • Google Cloud Professional Machine Learning Engineer • Certified Kubernetes Application Developer (CKAD) • HashiCorp Certified: Terraform Associate
Continuous Learning	• Angular Crash Course for Busy Developers, Udemy
Education	Bachelor of Engineering, Computer Science / Information Technology, Anna University, Chennai, India